The documentation and process conversion measures necessary to comply with this revision shall be completed by 23 November 2013.

INCH-POUND

MIL-PRF-19500/102B
23 August 2013
SUPERSEDING
MIL-S-19500/102A
29 December 1966

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTORS, NPN, SILICON, HIGH-POWER
TYPES 2N1016B, 2N1016C, AND 2N1016D JAN

Inactive for new design after 7 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a high-power, NPN, silicon transistor. One level of product assurance is provided for each device type.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. Unless otherwise specified T_A = +25°C.

Types	P_T T_C = +45°C (1)	$R_{\theta JC}$	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}
	W	°C/W	V dc	V dc	V dc	A dc	°C
2N1016B	150	0.7	100	25	100	7.5	-65 to +150
2N1016C	150	0.7	150	25	150	7.5	
2N1016D	150	0.7	200	25	200	7.5	

(1) Derate linearly 1.428 W/°C for T_C > +45°C. For temperature-power derating curves, see figure 2.

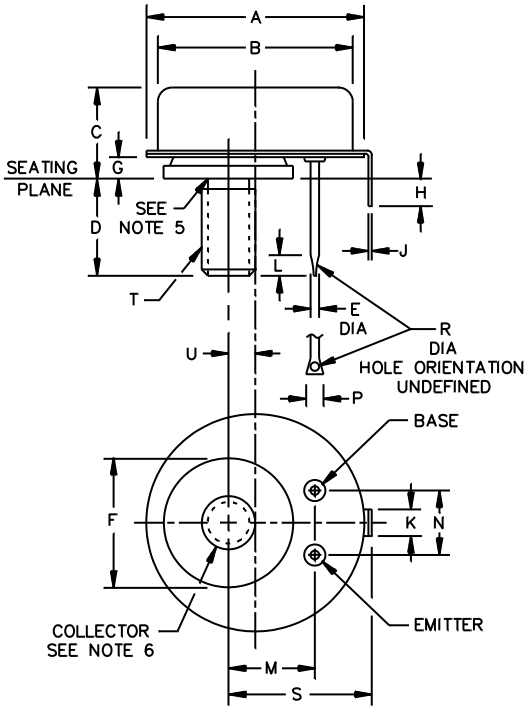
1.4 Primary electrical characteristics. Unless otherwise specified T_A = +25°C.

Limits	h_{FE} at V_{CE} = 4.0 V dc				$V_{CE(sat)}$ I_C = 5 A dc I_B = 1 A dc
	h_{FE1} I_C = 2.0 A dc	h_{FE2} I_C = 5.0 A dc	h_{fe} I_C = 5 A dc	f_{hfe} I_C = 5 A dc	
				kHz	V dc
Min	20	10	5	20	
Max	80	35			2.5

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	1.24	1.28	31.5	32.51	
B		1.130		28.7	
C	.50	.56	12.70	14.22	
D	.55	.58	13.97	14.73	
E	.045	.055	1.14	1.40	4
F	.73	.80	18.54	20.32	
G	.07	.14	1.78	3.56	
H	.13	.19	3.30	4.83	
J	.014	.024	0.36	0.61	
K	.14	.17	3.56	4.32	
L	.10	.14	2.54	3.56	
M	.48	.52	12.19	13.21	
N	.35	.40	8.89	10.16	
P	.09	.11	2.29	2.79	4
R	.050	.060	1.27	1.52	
S	.81	.85	20.57	21.59	
T					3
U	.14	.17	3.56	4.32	



NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. See NBS Handbook H28. 5/16 – 24 UNF-2A
4. Two leads.
5. Reference point for temperature measurement.
6. The collector shall be internally connected to the mounting base.

FIGURE 1. Physical dimensions.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3 and table I).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table II, group E, subgroup 4 herein.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.3.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.3.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
B3	1027	V_{CB} = 30 V dc, I_C = 2.39 A dc

4.3.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition A; weight = 5 pounds ±1 ounce, application time = 15 s Test condition D2; torque = 40 inch-pound; application time = 15 s. Test condition D1; torque = 5 in-oz; application time = 15 s; tubulated leads only.
C5	3151	$R_{\theta JC}$ (see 1.3).
C6	1026	1,000 hours, V_{CB} = 30 V dc, power and ambient temperature shall be applied to the device to achieve T_J = +150°C minimum, and minimum power dissipation of 75 percent of max rated P_T (see 1.3 herein); n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

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4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.2.3	$Z_{\theta JX}$			°C /W
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc	$V_{(BR)CEO}$			
2N1016B				100		V dc
2N1016C				150		V dc
2N1016D				200		V dc
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = -1.5 V dc,	I_{CEX1}			
2N1016B		V_{CE} = 100 V dc			1.0	mA dc
2N1016C		V_{CE} = 150 V dc			1.0	mA dc
2N1016D		V_{CE} = 200 V dc			1.0	mA dc
Collector to base cutoff current	3037	Bias condition D, V_{EB} = 1.5 V dc,	I_{CBO}			
2N1016B		V_{CB} = 100 V dc			1.0	mA dc
2N1016C		V_{CB} = 150 V dc			1.0	mA dc
2N1016D		V_{CB} = 200 V dc			1.0	mA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 25 V dc,	I_{EBO1}		1.0	mA dc
Collector to emitter voltage (saturated)	3071	I_C = 5 A dc, I_B = 1 A dc	$V_{CE(sat)1}$		2.5	V dc
Base emitter voltage (saturated)	3066	Test condition A, I_C = 5 A dc, I_B = 1 A dc	$V_{BE(sat)1}$		4.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 2 A dc	h_{FE1}	20	80	
Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 5 A dc	h_{FE2}	10	35	
Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 7.5 A dc	h_{FE3}	6	20	

See footnote at end of table.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - continued						
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 4 V dc, I_C = 5 A dc	h_{fe}	5		
Small-signal short- circuit forward-current transfer ratio cutoff frequency	3301	V_{CE} = 4 V dc, I_C = 5 A dc	f_{hfe}	20		kHz
Subgroup 3						
High temperature operation		T_C = +150°				
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 25 V dc	I_{EBO2}		10	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{EB} = -1.5 V dc	I_{CEX2}			
2N1016B		V_{CE} = 100 V dc			10	mA dc
2N1016C		V_{CE} = 150 V dc			10	mA dc
2N1016D		V_{CE} = 200 V dc			10	mA dc
Low-temperature operation		T_C = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 5 A dc	h_{FE4}	10	35	
Subgroup 4						
Pulse response		I_C = 5 A dc, V_{EB} = 6 V dc, V_{CE} = 12 V dc; I_{B1} and I_{B2} = 1.5 A dc (See figure 2)	$t_d + t_r$		10	µs
Pulse response		I_C = 5 A dc, V_{EB} = 6 V dc, V_{CE} = 12 V dc; I_{B1} and I_{B2} = 1.5 A dc (See figure 3)	$t_s + t_f$		15	µs
Subgroups 5, 6, and 7						
Not applicable						

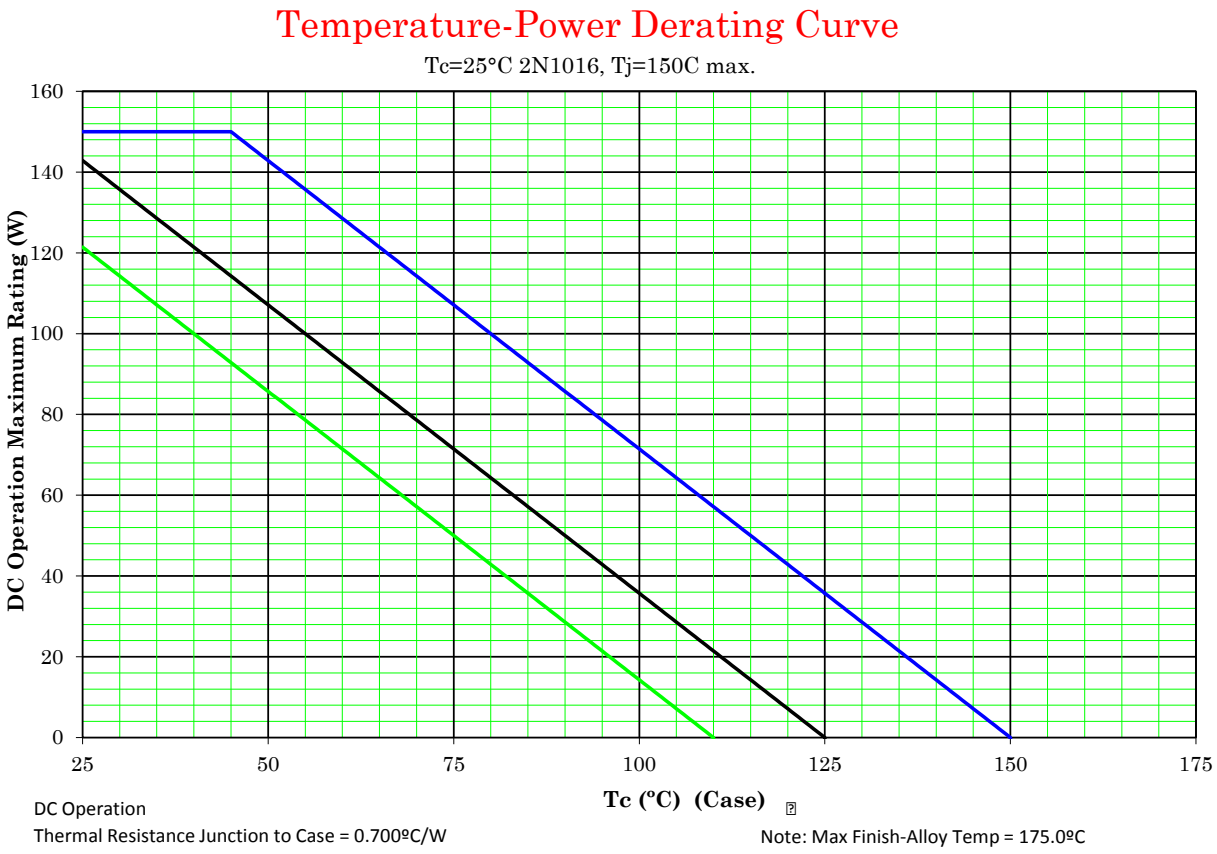
1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) - for qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Steady-state operating life	1026	V_{CB} = 30 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of 100°C.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves	3131	See MIL-PRF-19500, table E-IX, group E, subgroup 4.	
Subgroup 5			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq +150°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curves chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 2. Derating for all devices.

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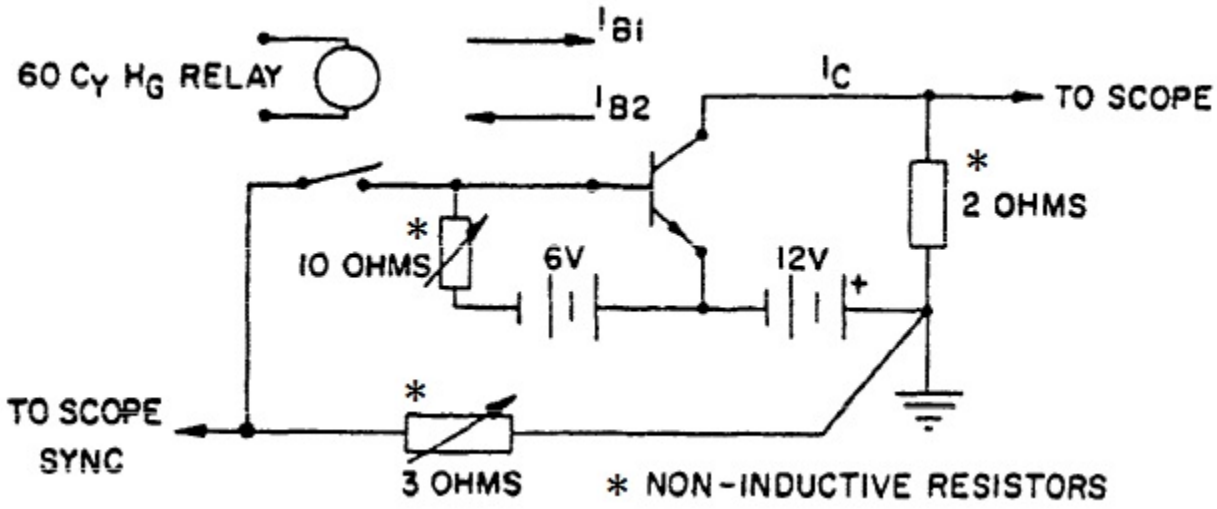


FIGURE 3. Pulse response test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil .

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2013-069)

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